Egger's test

Std_Eff	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
slope	2.754683	.1588469	17.34	0.000	2.431867	3.077498
bias	.1234977	.1484605	0.83	0.411	-.1782103	.4252056

Supplementary figure 1: Eggers's test for detection of publication bias for studies included in hypertension.

Egger's test

Std_Eff	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
slope	2.136301	.075485	28.30	0.000	1.974402	2.2982
bias	.6232349	.0960689	6.49	0.000	.4171876	.8292822

Supplementary figure 2: Eggers's test for detection of publication bias for studies included in overweight/obesity.

Egger's test

Std_Eff	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
slope	.6438857	.1736983	3.71	0.003	.2654296	1.022342
bias	1.377838	.2219967	6.21	0.000	.8941491	1.861528

Supplementary figure 3: Eggers's test for detection of publication bias for studies included in impaired glucose tolerance.

Egger's test

Std_Eff	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
slope	1.37992	.1365355	10.11	0.000	1.045829	1.71401
bias	1.060526	.2416536	4.39	0.005	.4692211	1.651831

Supplementary figure 4: Eggers's test for detection of publication bias for studies included in metabolic syndrome.

```
. metatrim pre sep, reffect graph funnel

Note: default data input format (theta, se_theta) assumed.

Meta-analysis

       |  Pooled      95% CI          Asymptotic      No. of
Method |     Est   Lower   Upper  z_value  p_value   studies
-------+-----------------------------------------------------
Fixed  |  13.384  12.999  13.769   68.133    0.000      16
Random |  23.975  19.938  28.012   11.641    0.000

Test for heterogeneity: Q= 1220.870 on 15 degrees of freedom (p= 0.000)
Moment-based estimate of between studies variance = 64.187

Trimming estimator: Linear
Meta-analysis type: Random-effects model

iteration |  estimate    Tn    # to trim     diff
----------+-----------------------------------------
    1     |   23.975     64          0         136
    2     |   23.975     64          0           0

Note: no trimming performed; data unchanged

Filled
Meta-analysis

       |  Pooled      95% CI          Asymptotic      No. of
Method |     Est   Lower   Upper  z_value  p_value   studies
-------+-----------------------------------------------------
Fixed  |  13.384  12.999  13.769   68.133    0.000      16
Random |  23.975  19.938  28.012   11.641    0.000

Test for heterogeneity: Q= 1220.870 on 15 degrees of freedom (p= 0.000)
Moment-based estimate of between studies variance = 64.187
```

Supplementary figure 5: Trim-and-fill analysis for the prevalence of overweight/obesity among adults in Ethiopia.

```
. metatrim pre sep, reffect graph funnel

Note: default data input format (theta, se_theta) assumed.

Meta-analysis

       |  Pooled      95% CI          Asymptotic      No. of
Method |     Est   Lower   Upper  z_value  p_value   studies
-------+-----------------------------------------------------
Fixed  |   6.235   5.825   6.644   29.814    0.000      14
Random |  12.426   8.753  16.100    6.630    0.000

Test for heterogeneity: Q= 764.415 on 13 degrees of freedom (p= 0.000)
Moment-based estimate of between studies variance = 47.131

Trimming estimator: Linear
Meta-analysis type: Random-effects model

iteration |  estimate    Tn    # to trim     diff
----------+-----------------------------------------
    1     |   12.426     60          1         105
    2     |   11.762     66          2          12
    3     |   10.877     73          3          14
    4     |   10.435     74          3           2
    5     |   10.435     74          3           0

Filled
Meta-analysis

       |  Pooled      95% CI          Asymptotic      No. of
Method |     Est   Lower   Upper  z_value  p_value   studies
-------+-----------------------------------------------------
Fixed  |   5.808   5.413   6.203   28.822    0.000      17
Random |  10.467   7.201  13.733    6.281    0.000

Test for heterogeneity: Q= 831.028 on 16 degrees of freedom (p= 0.000)
Moment-based estimate of between studies variance = 44.962
```

Supplementary figure 6: Trim-and-fill analysis for the prevalence of impaired glucose tolerance among adults in Ethiopia.

```
. metatrim pre sep, reffect graph funnel

Note: default data input format (theta, se_theta) assumed.

Meta-analysis

       |  Pooled      95% CI         Asymptotic      No. of
Method |     Est   Lower   Upper  z_value  p_value   studies
-------+-----------------------------------------------------
Fixed  |   7.055   6.663   7.446   35.317    0.000       8
Random |  14.708   9.820  19.596    5.897    0.000

Test for heterogeneity: Q= 500.276 on 7 degrees of freedom (p= 0.000)
Moment-based estimate of between studies variance = 47.566

Trimming estimator: Linear
Meta-analysis type: Random-effects model

iteration |  estimate    Tn    # to trim     diff
----------+-----------------------------------
    1     |   14.708     21         1          36
    2     |   13.294     22         1           2
    3     |   13.294     22         1           0

Filled
Meta-analysis

       |  Pooled      95% CI         Asymptotic      No. of
Method |     Est   Lower   Upper  z_value  p_value   studies
-------+-----------------------------------------------------
Fixed  |   7.025   6.634   7.415   35.248    0.000       9
Random |  13.298   8.682  17.915    5.646    0.000

Test for heterogeneity: Q= 505.145 on 8 degrees of freedom (p= 0.000)
Moment-based estimate of between studies variance = 47.100
```

Supplementary figure 7: Trim-and-fill analysis for the prevalence of metabolic syndrome among adults in Ethiopia.